

Jaime Alonso

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EDUCATION

Georgia Institute of Technology

May 2028

Bachelor of Science in Computer Science; Threads: Intelligence and Info Internetworks

Global Leadership LLC, Faculty Honors, GPA: 4.0

Certification: SEPE - DI - Industrial Applications of Machine Learning and Artificial Intelligence at Ironhack

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, C, JavaScript, HTML/CSS

ML/Data Frameworks: Scikit-learn, Pytorch, Pandas, NumPy, Streamlit, Jupyter Notebook, Matplotlib, XGBoost, HuggingFace

Developer Tools: Git, GitHub Actions, VS Code, PyCharm, Kobo Toolbox, Django

Cloud & AI: Google Cloud Platform, Gemini AI API Integration, PythonAnywhere, MySQL, Groq

EXPERIENCE

VIP Research Assistant at Thad Starner's Lab (Auto Robotics & VR Training in Construction)

Aug. 2025 – Present

Impact: Enabled ML-informed safety training by analyzing human computer interaction data from VR simulations

- Implemented instrumented VR simulation pipelines in Unity with event-driven logging systems to capture structured interaction signals for 5+ situations (hazard proximity, PPE compliance, user behavior)
- Processed and analyzed simulation data using Python (NumPy, Pandas, SciPy) to decrease safety risk by 32%
- Designed modular, script-driven simulation architectures and managed Git-based collaborative workflows (branching, merges, conflict resolution) to ensure reproducible, version-controlled analysis to 10+ researchers

Undergraduate Research Assistant at Georgia Institute of Technology, Gleason Lab

Jan. 2025 – Sep. 2025

Impact: Advanced healthcare research capabilities and data quality standards

- Applied ML models to analyze dataset of 6000+ patients to identify CPD risk factors affecting 65% of Ethiopian women
- Built binary classification models (Logistic Regression, Random Forest, and XGBoost) with 97.25% precision
- Processed and validated 400+ data entries in Kobo Toolbox, reducing data collection errors and expanding database archives
- Performed double data entry on 50+ records weekly, maintaining 99%+ accuracy through error checking protocols

Information Technology and Machine Learning Intern at MIT EI

July 2023 – Aug. 2023

Impact: Supported renewable energy research and improved team communication standards

- Accelerated renewable energy integration through analysis of 2+ years of ERCOT grid data by identifying distribution patterns that enabled strategic development of AI-driven grid optimization tool, GenX, for a 6-person team of MBA and PhD researchers
- Enhanced research team productivity by delivering daily presentations to senior researchers, establishing improved data delivery methodologies and best practices for knowledge transfer across disciplines

PROGRAMMING PROJECTS

LLM-Based News Intelligence Alerting System | Python, Groq API Llama 3.3, RSS, SMTP, GitHub Actions

July 2025 – Dec. 2025

- Developed a cloud-based Python automated personalized news report to 10+ email recipients daily with 100% uptime
- Designed an LLM-driven RSS filtering pipeline using semantic relevance classification for contextual content selection
- Implemented constrained prompt-based inference with Llama 3.3 to stabilize relevance decisions across 5+ domains
- Achieved zero-maintenance operation using serverless GitHub Actions with cron scheduling, mastering DevOps principles

Soccer League Explorer and Challenger | Gemini AI, Football-data.org API, Python, Streamlit, GitHub

Nov. 2024 – Dec. 2024

- Created a Streamlit webapp displaying top European Leagues with 95+ teams and real-time match data
- Strengthened API integration and AI prompt engineering with Google's Gemini AI processing 5+ API calls with 99%+ accuracy
- Optimized application performance to handle concurrent API requests and deployed via GitHub for real-time collaboration

La Ingeniería en tus Manos Project at Madrid Polytechnic University | C++, MIT App Inventor, Arduino

Oct. 2022 – June 2023

- Built 0.65-watt solar PV system with sun tracking capabilities from scratch, gaining hands-on hardware experience
- Led 4-member team as software developer captain, developing project management and technical leadership skills
- Created a full-stack web application using MIT App Inventor to enable Bluetooth connectivity and data gathering for 2+ users
- Gained valuable hardware skills with Arduino like welding and circuitry for 15+ electronic components

LEADERSHIP AND COMMUNITY ENGAGEMENT

Ramblin Rocket Club - Deputy Lead, Software and Simulations Teams

Student Government Association - Computer Science Representative and IT Committee member

GT Global Leadership LLC - Peer Mentor

GT Society of Hispanic Professional Engineers - Active Member

Interests: Sports enthusiast (Real Madrid, Alcaraz, Doncic), exploring cultures and cuisines, spending time with family and friends

Spoken Languages: English (Native), Spanish (Native), French (Fluent)